

$$\begin{aligned}
& C_{\phi}^{(22)} \quad (21) \rightarrow \frac{1}{16\pi^2} \left(-\mathbf{g}_1^6 \mathbf{m}_1^2 \tilde{\mu}^2 \text{LF}_{3,3,-1} [\mathbf{m}_1, \tilde{\mu}] - 3 \mathbf{g}_2^6 \mathbf{m}_2^2 \tilde{\mu}^2 \text{LF}_{3,3,-1} [\mathbf{m}_2, \tilde{\mu}] - \right. \\
& \mathbf{m}_1 \mathbf{g}_1^4 \mathbf{g}_2^2 \tilde{\mu}^2 (\mathbf{m}_1 + 2 \mathbf{m}_2) \text{LF}_{3,2,1,-1} [\tilde{\mu}, \mathbf{m}_1, \mathbf{m}_2] - \mathbf{m}_2 \mathbf{g}_1^2 \mathbf{g}_2^4 \tilde{\mu}^2 (2 \mathbf{m}_1 + \mathbf{m}_2) \text{LF}_{3,2,1,-1} [\tilde{\mu}, \mathbf{m}_2, \mathbf{m}_1] - \\
& \frac{1}{2} \mathbf{g}_1^2 \tilde{\mu}^2 y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} \text{LF}_{2,1,1,1,0} [\mathbf{m}_d^{\text{r}}, \mathbf{m}_d^{\text{t}}, \mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}] - \\
& \frac{3}{8} \tilde{\mu}^2 y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} (\mathbf{g}_1^2 + \mathbf{g}_2^2) \text{LF}_{2,2,1,1,-1} [\mathbf{m}_d^{\text{r}}, \mathbf{m}_q^{\text{p}}, \mathbf{m}_d^{\text{t}}, \mathbf{m}_q^{\text{s}}] + \\
& \frac{3}{8} \tilde{\mu}^2 y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} (\mathbf{g}_1^2 + \mathbf{g}_2^2) \text{LF}_{2,2,1,1,-1} [\mathbf{m}_d^{\text{r}}, \mathbf{m}_q^{\text{s}}, \mathbf{m}_d^{\text{t}}, \mathbf{m}_q^{\text{p}}] - \\
& \frac{1}{2} \mathbf{g}_1^2 \tilde{\mu}^2 y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} \text{LF}_{2,1,1,1,0} [\mathbf{m}_e^{\text{r}}, \mathbf{m}_e^{\text{t}}, \mathbf{m}_l^{\text{p}}, \mathbf{m}_l^{\text{s}}] - \\
& \frac{1}{8} \tilde{\mu}^2 y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} (\mathbf{g}_1^2 + \mathbf{g}_2^2) \text{LF}_{2,2,1,1,-1} [\mathbf{m}_e^{\text{r}}, \mathbf{m}_l^{\text{p}}, \mathbf{m}_e^{\text{t}}, \mathbf{m}_l^{\text{s}}] + \\
& \frac{1}{8} \tilde{\mu}^2 y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} (\mathbf{g}_1^2 + \mathbf{g}_2^2) \text{LF}_{2,2,1,1,-1} [\mathbf{m}_e^{\text{r}}, \mathbf{m}_l^{\text{s}}, \mathbf{m}_e^{\text{t}}, \mathbf{m}_l^{\text{p}}] + \\
& \frac{1}{4} \tilde{\mu}^2 y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} (\mathbf{g}_1^2 - \mathbf{g}_2^2) \text{LF}_{2,1,1,1,0} [\mathbf{m}_l^{\text{s}}, \mathbf{m}_e^{\text{r}}, \mathbf{m}_e^{\text{t}}, \mathbf{m}_l^{\text{p}}] - \\
& \frac{1}{4} \tilde{\mu}^2 y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} (\mathbf{g}_1^2 - 3 \mathbf{g}_2^2) \text{LF}_{2,1,1,1,0} [\mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}, \mathbf{m}_u^{\text{r}}, \mathbf{m}_u^{\text{t}}] + \\
& \frac{3}{8} \tilde{\mu}^2 y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} (\mathbf{g}_1^2 + \mathbf{g}_2^2) \text{LF}_{2,2,1,1,-1} [\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{s}}, \mathbf{m}_u^{\text{t}}] - \\
& \frac{3}{8} \tilde{\mu}^2 y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} (\mathbf{g}_1^2 + \mathbf{g}_2^2) \text{LF}_{2,2,1,1,-1} [\mathbf{m}_q^{\text{p}}, \mathbf{m}_u^{\text{t}}, \mathbf{m}_q^{\text{s}}, \mathbf{m}_u^{\text{r}}] - \\
& \frac{1}{4} \tilde{\mu}^2 y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} (\mathbf{g}_1^2 + 3 \mathbf{g}_2^2) \text{LF}_{2,1,1,1,0} [\mathbf{m}_q^{\text{s}}, \mathbf{m}_d^{\text{r}}, \mathbf{m}_d^{\text{t}}, \mathbf{m}_q^{\text{p}}] + \\
& \mathbf{g}_1^2 \tilde{\mu}^2 y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{2,1,1,1,0} [\mathbf{m}_u^{\text{r}}, \mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}, \mathbf{m}_u^{\text{t}}] + \\
& 3 \tilde{\mu}^2 \overline{y_d^{\text{uv}}} y_d^{\text{pt}} y_d^{\text{sv}} y_d^{\text{ur}} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} \text{LF}_{1,1,1,1,1,0} [\mathbf{m}_d^{\text{r}}, \mathbf{m}_d^{\text{t}}, \mathbf{m}_d^{\text{v}}, \mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}] + \\
& 3 \tilde{\mu}^2 \overline{y_d^{\text{uv}}} y_d^{\text{pt}} y_d^{\text{sv}} y_d^{\text{ur}} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} \text{LF}_{1,1,1,1,1,0} [\mathbf{m}_d^{\text{r}}, \mathbf{m}_d^{\text{t}}, \mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}, \mathbf{m}_q^{\text{u}}] + \\
& \tilde{\mu}^2 \overline{y_e^{\text{uv}}} y_e^{\text{pt}} y_e^{\text{sv}} y_e^{\text{ur}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} \text{LF}_{1,1,1,1,1,0} [\mathbf{m}_e^{\text{r}}, \mathbf{m}_e^{\text{t}}, \mathbf{m}_e^{\text{v}}, \mathbf{m}_l^{\text{p}}, \mathbf{m}_l^{\text{s}}] + \\
& \tilde{\mu}^2 \overline{y_e^{\text{uv}}} y_e^{\text{pt}} y_e^{\text{sv}} y_e^{\text{ur}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} \text{LF}_{1,1,1,1,1,0} [\mathbf{m}_e^{\text{r}}, \mathbf{m}_e^{\text{t}}, \mathbf{m}_l^{\text{p}}, \mathbf{m}_l^{\text{s}}, \mathbf{m}_l^{\text{u}}] + \\
& 3 \tilde{\mu}^2 y_d^{\text{sv}} y_d^{\text{ur}} \overline{a_d^{\text{pr}}} \overline{a_d^{\text{st}}} \overline{a_d^{\text{uv}}} a_d^{\text{pt}} \text{LF}_{1,1,1,1,1,1,0} [\mathbf{m}_d^{\text{r}}, \mathbf{m}_d^{\text{t}}, \mathbf{m}_d^{\text{v}}, \mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}, \mathbf{m}_q^{\text{u}}] + \\
& \tilde{\mu}^2 y_e^{\text{sv}} y_e^{\text{ur}} \overline{a_e^{\text{pr}}} \overline{a_e^{\text{st}}} \overline{a_e^{\text{uv}}} a_e^{\text{pt}} \text{LF}_{1,1,1,1,1,1,0} [\mathbf{m}_e^{\text{r}}, \mathbf{m}_e^{\text{t}}, \mathbf{m}_e^{\text{v}}, \mathbf{m}_l^{\text{p}}, \mathbf{m}_l^{\text{s}}, \mathbf{m}_l^{\text{u}}] + \\
& \left. 3 \tilde{\mu}^4 \overline{y_u^{\text{uv}}} y_u^{\text{pt}} y_u^{\text{sv}} y_u^{\text{ur}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} \text{LF}_{1,1,1,1,1,1,0} [\mathbf{m}_q^{\text{p}}, \mathbf{m}_q^{\text{s}}, \mathbf{m}_q^{\text{u}}, \mathbf{m}_u^{\text{r}}, \mathbf{m}_u^{\text{t}}, \mathbf{m}_u^{\text{v}}] \right)
\end{aligned}$$